

IIMS Sensor Processing Project

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1 Objective

You will work with real-world marine sensor data to process, analyze, and interpret measurements in the context of their acquisition. Your task is to tell a compelling story that is supported by your data analysis (i.e. what patterns, anomalies, or environmental insights can you extract?). Additionally, you should explore how your findings relate to the sensor's deployment context

2 Structure

Students will work in groups of 3 (minimum 2). The final deliverables of this project will be:

- A small report (around 3 pages of text not counting figures) as PDF file, which details your methodology, analysis and interpretation. The submission must also include any code that was used to process the data to make your work transparent and reproducible.
- A small presentation not more than 7 min about your findings. There will be an additional 3 minutes for questions from the audience. The main aim of the presentation is to make sure everyone in the group has understood the data, methodology and interpretation that was submitted in the report.

Report deadline: 15.06.2026.

Presentation date: 16.06.2026.

2.1 grading

The project will be graded based on three key criteria:

- How well your processing approach fits the data and how thoroughly your analysis is supported by your methods and results
- The reproducibility of your code

- The quality of your presentation, including your explanations and responses to questions

The resulting grade will count for **33** % of your final grade for the course.

3 Data

Datasets from different sensors will be available for your group to choose from for analysis. The dataset types are the following:

1. Passive acoustic data from a hydrophone [[icListen HF from Ocean Sonics](#)] deployed in [Unisee](#), while an uncrewed surface vehicle (USV) [[USV Otter from Maritime Robotics](#)] was driving around nearby.
 - The task would be to identify the USV in the sound recordings and to explore if there is something else that can be derived from the dataset
2. Single beam echo-sounder [[ISA500 from Impact Subsea](#)] data in combination with GNSS information recorded from an USV [[USV Otter from Maritime Robotics](#)] driving around in the [Unisee](#).
 - The task would be to generate a sparse height-map of the seafloor. The distance from the seafloor can be derived from the single beam echo sounder measurements and combined with GNSS fixes to get the measurement location.
3. Camera data from an USV [[USV Otter from Maritime Robotics](#)] driving around in the [Unisee](#).
 - A possible task would be to use the camera to identify "no-go" areas for the USV based on pre-defined rules (e.g. possible nesting spots for birds, swimming areas for humans, etc.).
 - Another possible task would be to derive navigation information from the camera data.
4. Salinity data gathered by an ocean glider [[SLOCUM G3 by Teledyne](#)] in the Atlantic Ocean.
 - The task would be to identify the structure of the density distribution of the part of the ocean traversed by the glider.

Note: This dataset will require background knowledge in physical oceanography for contextualization and interpretation.

Additionally, for each dataset we will make **metadata** available describing the

context of the data-acquisition.

Tools to use: You can analyze the data with the tools of your choice as long as you can make the processing transparent and reproducible (i.e. provide code or worksheets).

4 Processing

The expected processing, analysis and interpretation can be divided into the following steps:

1. Data inspection and context:
 - understand the device with which the data was gathered
 - understand the context in which the data was gathered
 - visualize and inspect the raw data
2. Data processing:
 - preprocessing (e.g. handling of missing data, outliers and possible corrections)
 - filtering and smoothing where needed (i.e. handle measurement noise and biases)
 - feature extraction where needed (i.e. derive secondary variables where relevant)
 - combining data where relevant
 - data validation where possible (i.e. are there other available data sources which support your processing?)
3. Analysis
 - explore your data by plotting key variables
 - search for and identify patterns where relevant
 - statistics where relevant
 - interpret the data within the given context
 - consider and discuss limitations inherent in the data and in the processing approach
4. Reporting
 - structure your story: **Context / Problem** $\rightarrow$ **Data** $\rightarrow$ **Methods** $\rightarrow$ **Results** $\rightarrow$ **Interpretation / Discussion**
 - consider how to best visualize the data to report your findings
 - report your methodology in a manner that makes your work transparent and reproducible

- write your report and prepare your presentation following the above structure